

# Intro to ECO 3253

## Economics of Public and Social Issues

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ECO3253, UTSA

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What is this course about?

# This course

Welcome to **ECO 3253!**

Big picture:

- ▶ How to use data to understand and solve current and important economic problems!
- ▶ Get a sense of research frontier in applied economics and social science (equality of opportunity and mobility, education, innovation and entrepreneurship, health care, climate change, and crime, etc)
- ▶ How will we get there? By doing 3 things:
  1. covering the topics with a focus towards using data that can help answer these questions
  2. understanding the intuition for how to use data to answer these questions (basics of statistical analysis)
  3. using computational tools to help us with the statistical analysis (basics of R)

# About me

Let me briefly tell you about me:

- ▶ Colombian
- ▶ PhD in Economics at [Duke University](#) in North Carolina
- ▶ Masters in Economics at [Université Catholique de Louvain](#) in Belgium
- ▶ Undergraduate in (you guessed it!) Economics at [Universidad Nacional de Colombia](#).
- ▶ Applied (micro)economist:
  - ▶ 'micro': focus on specific markets (housing and media)
  - ▶ 'applied': I use data all the time

# Website

We will have a cool website with the materials covered in class:

[https://jrm87.github.io/ECO3253\\_fall2026/](https://jrm87.github.io/ECO3253_fall2026/)

Info on Canvas. Write it down. Keep it in mind!

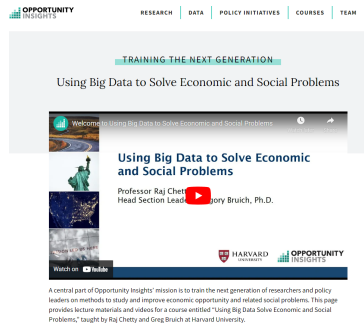
# Topics to be covered - 1

Lectures based on the [course](#) by [Prof. Raj Chetty](#).

R: Chetty



Opportunity Atlas Course

A screenshot of the Opportunity Insights website. The header includes the logo and navigation links: RESEARCH, DATA, POLICY INITIATIVES, COURSES, and TEAM. The main content area features the heading "TRAINING THE NEXT GENERATION" and the subtitle "Using Big Data to Solve Economic and Social Problems". Below this is a video player with the title "Using Big Data to Solve Economic and Social Problems" and the text "Professor Raj Chetty, Head Section Leader" and "Gregory Bruch, Ph.D.". The video player has a "Watch on YouTube" button. At the bottom, there are logos for Harvard University and Opportunity Insights, and a paragraph of text: "A central part of Opportunity Insights' mission is to train the next generation of researchers and policy leaders on methods to study and improve economic opportunity and related social problems. This page provides lecture materials and videos for a course entitled 'Using Big Data Solve Economic and Social Problems,' taught by Raj Chetty and Greg Bruch at Harvard University."

## Topics to be covered - 2

The plan for what we will covered in the semester includes:

- ▶ Geography of Upward Mobility in America
- ▶ Causal Effects of Neighborhoods and Characteristics of High-Mobility Areas
- ▶ Historical and International Evidence on the Drivers of Inequality and Mobility
- ▶ Upward Mobility, Innovation, and Growth
- ▶ Higher Education and Upward Mobility
- ▶ Primary education
- ▶ Teachers and Charter Schools
- ▶ Racial Disparities in Economic Opportunity
- ▶ Improving judicial decisions
- ▶ Immigration
- ▶ Political Economy
- ▶ Income taxation
- ▶ Savings and wealth
- ▶ Housing markets and COVID
- ▶ Intro to air and water pollution, and externalities

# Data-driven course

- ▶ Data driven, not just theoretical
- ▶ We will work with real world data, and we will try to make sense of it.
- ▶ Economics is much more empirical now than a few years ago

# Number of data-driven articles in leading journals

**Empirical (Data-Based) Articles in Leading Economics Journals, 1983-2011**

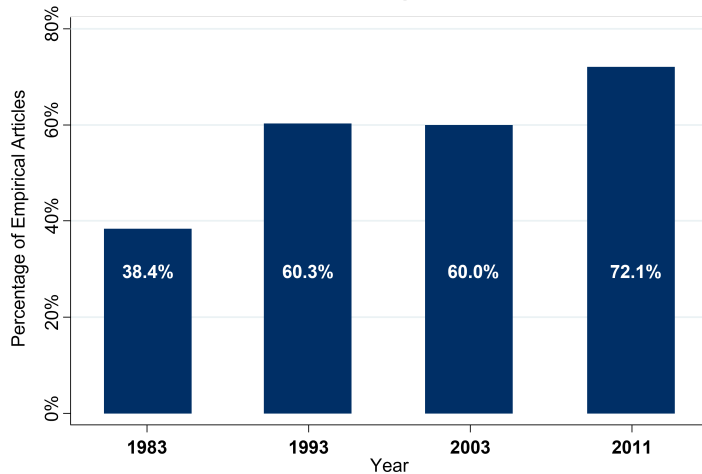


Figure 1: Source: Mameresh (JEL 2013)

# What do we need to understand data?

How can we know if going to school increases wages? Or if economic mobility is low or high?

- ▶ Framework (basic stats and intuition)
  - ▶ I will provide all the basic intuition for the empirical methods. These include:
    - ▶ Descriptive Data Analysis: correlation, regression
    - ▶ Experiments: randomization, non-compliance
    - ▶ Quasi-Experiments: regression discontinuity, difference-in-differences
    - ▶ Machine Learning: prediction, overfitting, cross-validation
- ▶ A powerful (and very cool) calculator:
  - ▶ **R** as our statistical software
  - ▶ Will follow parts of the book [ModernDive](#)
  - ▶ I will guide you through it - does not matter if you have no experience with 'programming'

## Ok, but why R?

- ▶ **R** is free and open source!
- ▶ **R** has a vibrant online community!
- ▶ **R** is very flexible and powerful — adaptable to nearly any task ( e.g., correlations, econometrics, spatial data analysis, machine learning, web scraping, data cleaning, website building, teaching.)
- ▶ Employers like **R** over alternatives

# R and RStudio

- ▶ We will talk more about R and RStudio in our next class.
- ▶ I will reach out with very simple instructions sometime this week.
- ▶ We will work with a **local installation** on your own laptop — both R and RStudio are **free**.
  - ▶ If you would rather try a free **Posit Cloud** account in the browser, there are notes on the course site. That is optional, and **there is no class invite this semester**.

# Economic Concepts

Back to the content! We will cover and use several economic concepts you probably learned before. We will see them in practice:

1. Effects of price incentives
2. Supply and demand
3. Competitive equilibrium
4. Adverse selection
5. Behavioral economics vs. rational models

# Grades

| Activity                           | Perc. Weight |
|------------------------------------|--------------|
| Projects 0–4                       | 12%          |
| Quizzes (weekly, 2 lowest dropped) | 12%          |
| Midterm                            | 23%          |
| Final                              | 23%          |
| Attendance and participation       | 30%          |
| <b>Total</b>                       | <b>100%</b>  |

# Quizzes

- ▶ **Short quiz at the start of class every Thursday** (~5 min), run through **iClicker**
  - ▶ It **opens when class starts and closes a few minutes later** — arrive on time
  - ▶ Covers the previous sessions and the assigned readings
- ▶ **No quiz** on Thursdays we do not meet (09/24, 10/08, Thanksgiving 11/26), and none on day one
- ▶ That gives **12 quizzes**. Your **2 lowest scores are dropped** — only your best 10 count
- ▶ Because the 2 drops already cover missed or bad days, there are **no make-up quizzes**  
*Miss the window and that quiz simply becomes one of your two drops.*

## iClicker — set this up now

We use **iClicker** every class: for the Thursday quizzes, and for short in-class polls on the material that count toward participation. It is **free for all UTSA students**.

1. Create an account at **iclicker.com** with your **UTSA email**
2. Install the **iClicker student app**, or use the browser version at [iclicker.com](https://iclicker.com)
3. Add this course — it should appear via Canvas, or use the **QR code / join code** I show in class
4. Bring a **charged device** to every class  
*Polls are graded for **answering and thinking**, not for being right. No device?  
Tell me — do not just skip it.*

# Projects

- ▶ A big part of the course: **4 projects** (plus a warm-up “project 0”)
- ▶ More involved than typical homeworks: real data, real problems, real code
- ▶ Submitted on **Canvas** by **11:59 pm** on the due date
- ▶ **How they are graded:**
  - ▶ Each on-time submission earns **full credit** (✓) toward the **Projects 0–4 (12%)** bucket — I do not line-by-line correct them
  - ▶ The **substance** of what you submitted (your reasoning, assumptions, code choices, interpretation) is evaluated separately, through the **Attendance & participation (30%)** bucket — see next slide
- ▶ You may use generative AI tools, but you are responsible for verifying outputs and being able to explain every line you submit
- ▶ Start early! Many new elements — give yourself time to try, fail, and experiment  
*Do not work on these projects the day before!*

## Participation: how the 30% works

This is a face-to-face course built around active discussion and frequent reference to the projects you submit.

- ▶ On any class day, I may **randomly call on you** to:
  - ▶ answer or expand on the topic we are covering in lecture, or
  - ▶ **explain a piece of your submitted project** (code logic, identification assumptions, interpretation of a result)
- ▶ If you prefer, you can “**call on a friend**” — a classmate volunteers to help expand the answer
- ▶ If you are **called and absent** without documentation, that missed opportunity counts toward the participation score (see UTSA policies for excused absences)

## What strong participation looks like

- ▶ **Preparation & relevance.** You come prepared, reference the assigned readings or methods, and tie comments back to your own submitted work
- ▶ **Reasoning & evidence.** You state your assumptions, defend your identification strategy, and recognize when an estimate or AI suggestion needs to be verified
- ▶ **Code fluency.** You can walk through key parts of your code (data wrangling, estimation, diagnostics), make a small on-the-spot edit, and explain *why* it addresses the question
- ▶ **Dialogue.** Respectful engagement, clarifying questions, willingness to call on a friend to extend a point

## Using AI in this course

# Does using AI actually help you learn?

New evidence: [Strömberg, Lei and Wu \(2026\)](#), *The Generative AI Learning Penalty*

- ▶ **26,811 Chinese students**, grades 7–12, followed for **30 months**
- ▶ AI use went from ~0% in 2022 to ~**80%** by June 2025, and students adopted it at different times — so we can compare a student **to herself**, before vs. after
- ▶ A **difference-in-differences** design — the method we learn in this course

# Does using AI actually help you learn?

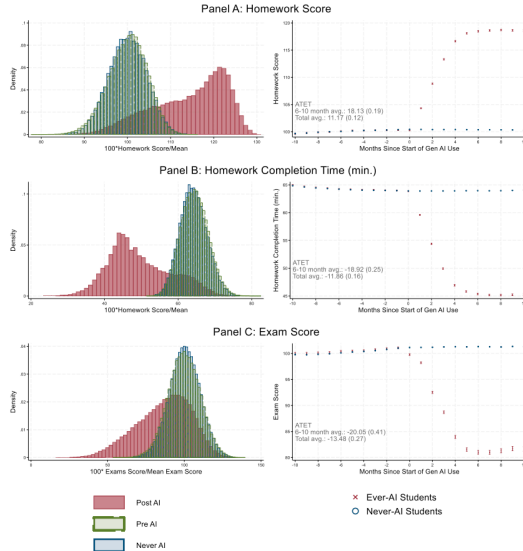
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A split between **doing the work** and **learning from it**:

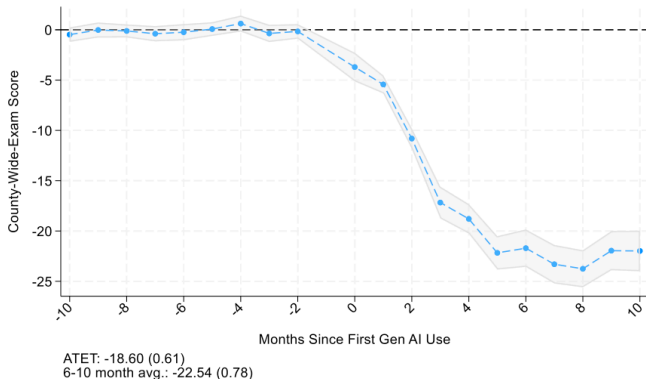
- ▶ **Homework got better**: scores **+18%**, time per assignment **64 → 45 min**
- ▶ **Learning got worse**: closed-book exam scores **−20%** within six months

# Homework up, learning down



Red = after AI; blue/green = before AI, or never used it.

# The damage compounds



- ▶ Flat **before** adoption, falling sharply **after** — that flat pre-trend is what makes this credible
- ▶ The **full** penalty takes about **two years** to appear — short studies miss it

## Who gets hurt, and how much

- ▶ High-stakes entrance exams: **−18%** (college) and **−24%** (high school)
- ▶ Worst in **social sciences (−27%)**, then **STEM (−22%)**, then languages (−9 to −17%)
- ▶ Hit hardest: **younger students, boys, and high-achieving students**

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**The mechanism.** About **81%** of AI users showed *homework outsourcing*: very fast completion, high homework scores, poor closed-book performance.

***But students who used AI and still spent as much time on the work as non-users scored the same on exams.***

The tool was not the problem. The **thinking time it replaced** was the problem.

## How I want you to use AI in this course

- ▶ Use AI to **check, explain, and debug** — not to produce what you hand in
- ▶ If it made the task faster, **spend the time you saved** on understanding it
- ▶ **If you cannot explain it, you did not learn it** — this is why I ask you to walk through your own code in class
- ▶ This is also why we have **weekly quizzes**: retrieval practice is what makes it stick

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Want the short version first? I left a two-minute summary from *The Economist* on Canvas: [AI and children's learning](#). The full paper is [here](#) and in the readings folder.